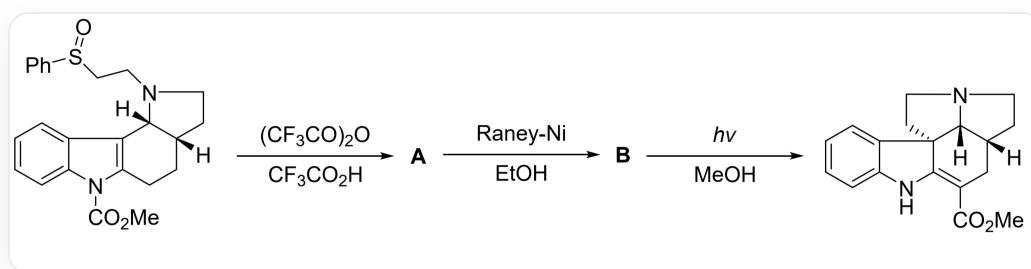


Question

Pummerer rearrangement typically refers to the chemical reaction of activating the oxygen atom of a sulfoxide with an anhydride, going through a sulfonium ion intermediate, and ultimately generating an α -functionalized sulfide. In the total synthesis of a certain natural product (as shown in the figure below), Pummerer rearrangement was used to construct a new quaternary carbon center.



The figure shows a multi-step reaction, which can be described as: [H]

[C@@]1(CCN(CCS(C2=CC=CC=C2)=O)[C@]13[H])CCC4=C3C5=CC=CC=C5N4C(OC)=O>

(CF₃CO)₂O, CF₃CO₂H>[A], [A]>Raney – Ni, EtOH>[B], [B]>hv, MeOH>[H]

[C@@]1(CC(C(OC)=O)=C2N([H])C3=CC=CC=C3[C@]24CCN5CC1)[C@]45[H] °Where [1]>'2'>[3] means compound 1 reacts under the conditions of 2 to obtain compound 3.

Which of the following statements about **A** and **B** are correct:

- A** has three stereochemical centers
- A** and **B** have the same number of rings
- A carbon-carbon double bond shift occurred during the generation of **B** from **A**
- [1,5]- σ migration occurred in **B** under light irradiation
- Compared with **B**, there is no generation or disappearance of stereochemical centers in the final product
- An imine structure exists in the intermediate of the formation of the final product from **B**

A. 1,2,4,5,6

B. 1,2,3,4,5,6

C. 5,6

D. 2,6

E. 2,3,6

F. 1,2,4,6

G. 1,3,4,5

H. 2,4,5

I. 3,4,6

J. All of the above options are incorrect or the answer is incomplete.

Answer

Correct Answer: C

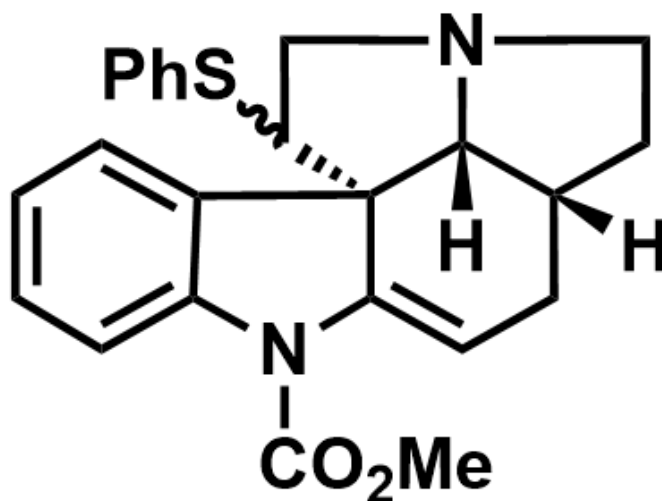
Detailed Explanation

In the process of the reactant generating **A**, firstly, the classical Pummerer rearrangement reaction occurs, that is, the oxygen atom of the sulfoxide nucleophilically attacks trifluoroacetic anhydride, resulting in a sulfonium ion, followed by 1,2-elimination to obtain a carbocation (or an onium ion containing a carbon-sulfur double bond) at the ortho position of the sulfide. Since the indole ring is relatively electron-rich, it can undergo an aromatic electrophilic substitution reaction. The number 3 position of the indole ring nucleophilically attacks the carbocation, constructing a five-membered ring system similar to the product, and generating an imine cation structure, and finally eliminating H^+ to obtain **A**, the structure is as follows:

CHECKPOINT

1 PTS

The number 3 position of the indole ring nucleophilically attacks the carbocation, constructing a five-membered ring system similar to the product



[H][C@@]1(CC=C([C@@](C2=CC=CC=C32)4C(CN5CC1)SC6=CC=CC=C6)N3C(OC)=O)[C@]45[H]

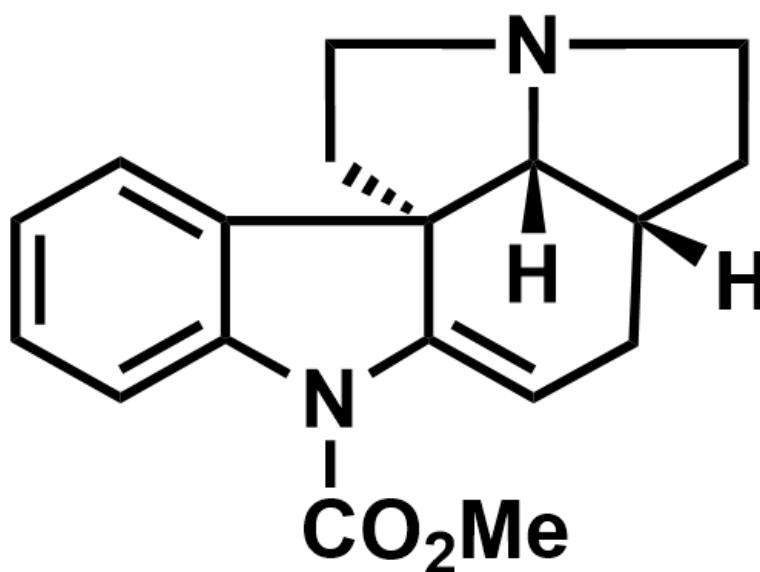
CHECKPOINT

1 PTS

The structure of **A** is [H][C@@]1(CC=C([C@@](C2=CC=CC=C32)4C(CN5CC1)SC6=CC=CC=C6)N3C(OC)=O)[C@]45[H]

So **A** has four stereochemical centers. Statement 1 is incorrect.

A is then reduced by Raney – Ni to cleave the carbon-sulfur bond, finally yielding one molecule of PhSH and **B**. The structure of **B** is as follows:



[H][C@@]1(CC=C([C@@](C2=CC=CC=C32)4CCN5CC1)N3C(OC)=O)[C@]45[H]

CHECKPOINT

1 PTS

Raney — Ni reduces the carbon-sulfur bond

CHECKPOINT

1 PTS

The structure of **B** is [H][C@@]1(CC=C([C@@](C2=CC=CC=C32)4CCN5CC1)N3C(OC)=O)[C@]45[H]

Because it is reduced, **B** has one less benzene ring than **A**, and no carbon-carbon double bond shift occurs in the process of **A** generating **B**. Statements 2 and 3 are incorrect.

CHECKPOINT

1 PTS

A has one more ring than **B**, and no carbon-carbon double bond shift occurs in the process of **A** generating **B**

Under illumination, the $-\text{COOMe}$ in **B** undergoes a $[1,3]-\sigma$ migration, yielding an imine intermediate, followed by imine-enamine tautomerization to give the final product. In this step, there is no creation or disappearance of stereochemical centers. Therefore, statements 4 is incorrect, and 5 and 6 are correct.

CHECKPOINT

1 PTS

B undergoes a $[1,3]-\sigma$ migration under illumination

CHECKPOINT

1 PTS

Compared with **B**, there is no creation or disappearance of stereochemical centers in the final product

CHECKPOINT

1 PTS

There is an imine structure in the intermediate of **B** generating the final product

Therefore, statements 5 and 6 are correct, and the answer is C.